

Vehicle Service Management System

INTRODUCTION

- In the modern digital era, vehicle service centers face challenges in managing customer requests, service bookings, and vehicle records using manual or semi-automated processes. Customers often experience difficulties such as lack of transparency in service status, delayed communication, and improper record maintenance. To overcome these issues, an efficient and centralized digital platform is required.
- The **Vehicle Service Management System** is a web-based application developed to automate and streamline the complete vehicle service workflow. This system provides an easy-to-use platform for users to register, log in, book vehicle services, and track service status, while administrators can manage bookings, approve service requests, and monitor overall operations. The project improves service efficiency, reduces manual work, and enhances customer satisfaction.

ABSTRACT

- The Vehicle Service Management System is designed to provide a complete solution for managing vehicle service operations digitally. The main objective of this project is to simplify the interaction between vehicle owners and service centers by providing an online service booking and management platform.
- Users can register, log in, add their vehicle details, book service appointments, and track the status of their bookings. Administrators can manage service requests, approve or reject bookings, update service status, and maintain service records. The system is developed using Spring Boot for backend processing, Thymeleaf with HTML and CSS for the frontend, and MySQL for database management. This project demonstrates a real-world application of full-stack development and serves as an effective capstone project.

What We Are Going to Build: Client Requirement

- The client requires a Vehicle Service Management System that enables efficient management of vehicle service operations. The system should support both user and admin roles with secure authentication and authorization.
- Client requirements include:
 - Users should be able to register and log in securely.
 - Users should be able to add vehicle details and book service appointments.
 - Users should be able to view and track their service booking status.
 - Admin should be able to view all service bookings.
 - Admin should be able to approve, update, or complete service requests.
 - The system should maintain proper service history and records.

What We Are Going to Build: Some Technical Terms

- Secure Login and Registration functionality.
- Role-based authentication for Admin and User.
- CRUD operations for vehicles, services, and bookings.
- Proper validation of user inputs.
- Exception handling for error-free execution.
- MVC (Model–View–Controller) architecture.
- Database connectivity using JPA and Hibernate.
- User-friendly UI with professional design.

What Are the Technologies and Tools We Are Going to Use?

- **Backend Framework:** Spring Boot (Java)
- **Frontend:** HTML, CSS, Thymeleaf
- **Programming Language:** Java (JDK 8+)
- **Database:** MySQL
- **ORM Tool:** Hibernate / Spring Data JPA
- **Build Tool:** Maven
- **IDE:** Spring Tool Suite (STS)
- **Web Server:** Apache Tomcat
- **Testing Tool:** JUnit

SYSTEM REQUIREMENTS

- **Software Requirements:**

- Backend: Spring Boot, Spring Data JPA, Hibernate
- Database Server: MySQL Server 8.0
- Web Server: Apache Tomcat
- IDE: Spring Tool Suite (STS)
- Browser: Google Chrome
- Testing: JUnit

- **Hardware Requirements:**

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Hard Disk: Minimum 20 GB

PROJECT MODULES

- Admin Module
- User Module
- Vehicle Module
- Service Booking Module

ADMIN MODULE

The Admin Module allows administrators to manage the entire system after successful login. Admin functionalities include viewing all service bookings, approving or rejecting service requests, updating service status, and managing service records. The admin plays a key role in maintaining smooth service operations.

USER MODULE

- The User Module allows vehicle owners to register and log in to the system. Users can add vehicle details, book service appointments, and track booking status such as Pending, Approved, or Completed. This module provides transparency and convenience to customers by enabling real-time service tracking.

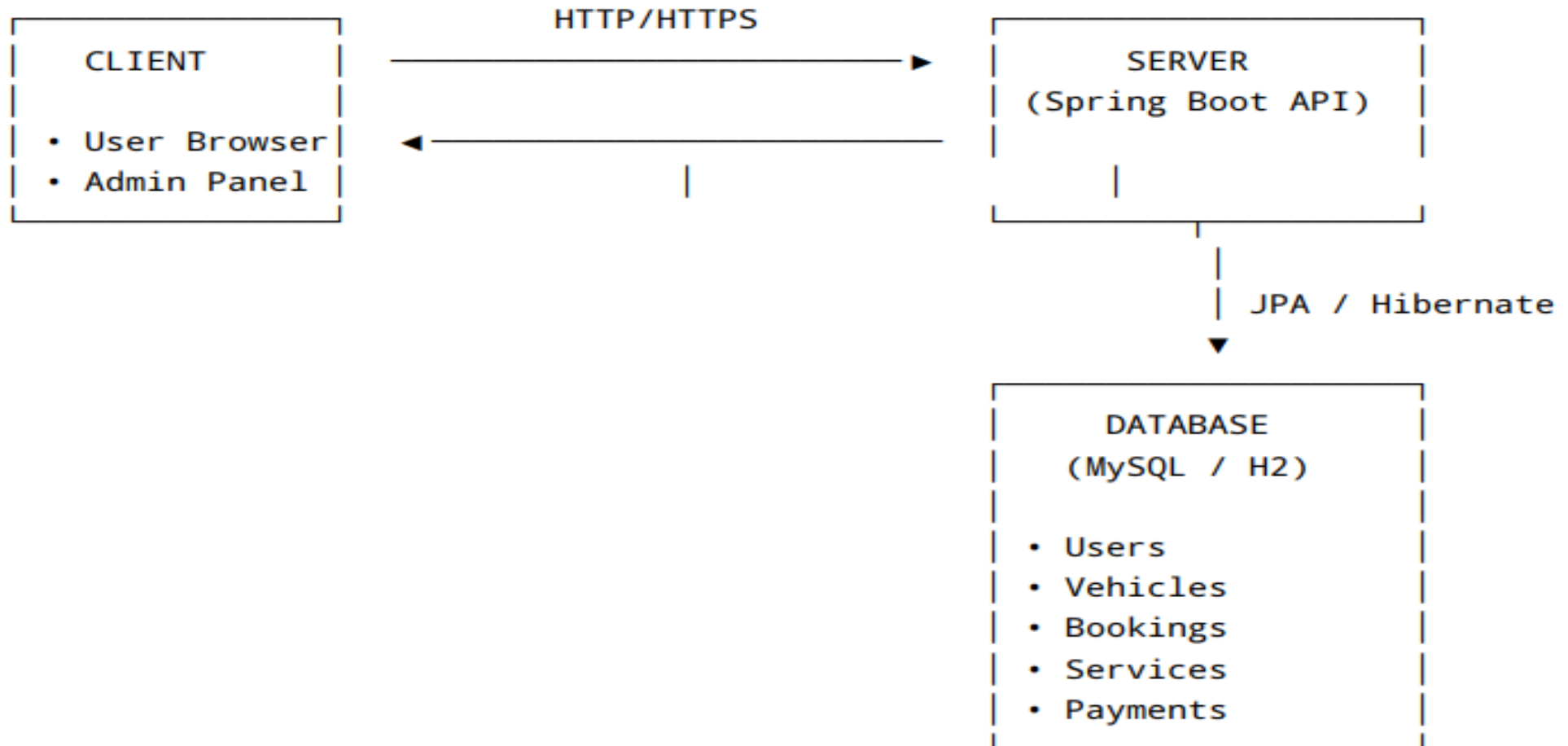
VEHICLE MODULE

- The Vehicle Module is responsible for managing vehicle-related information within the system. This module allows users to add and maintain details of their vehicles, which are required for booking service appointments. Each vehicle is uniquely associated with a registered user, ensuring proper identification and service history tracking.
- In this module, users can add vehicle details such as vehicle number, vehicle type, brand, model, and fuel type. Users can also view, update, or delete their vehicle information whenever required. This module helps in maintaining accurate records of vehicles and avoids confusion during service processing. The Vehicle Module plays an important role in linking vehicles with service bookings and maintaining complete service history for future reference.

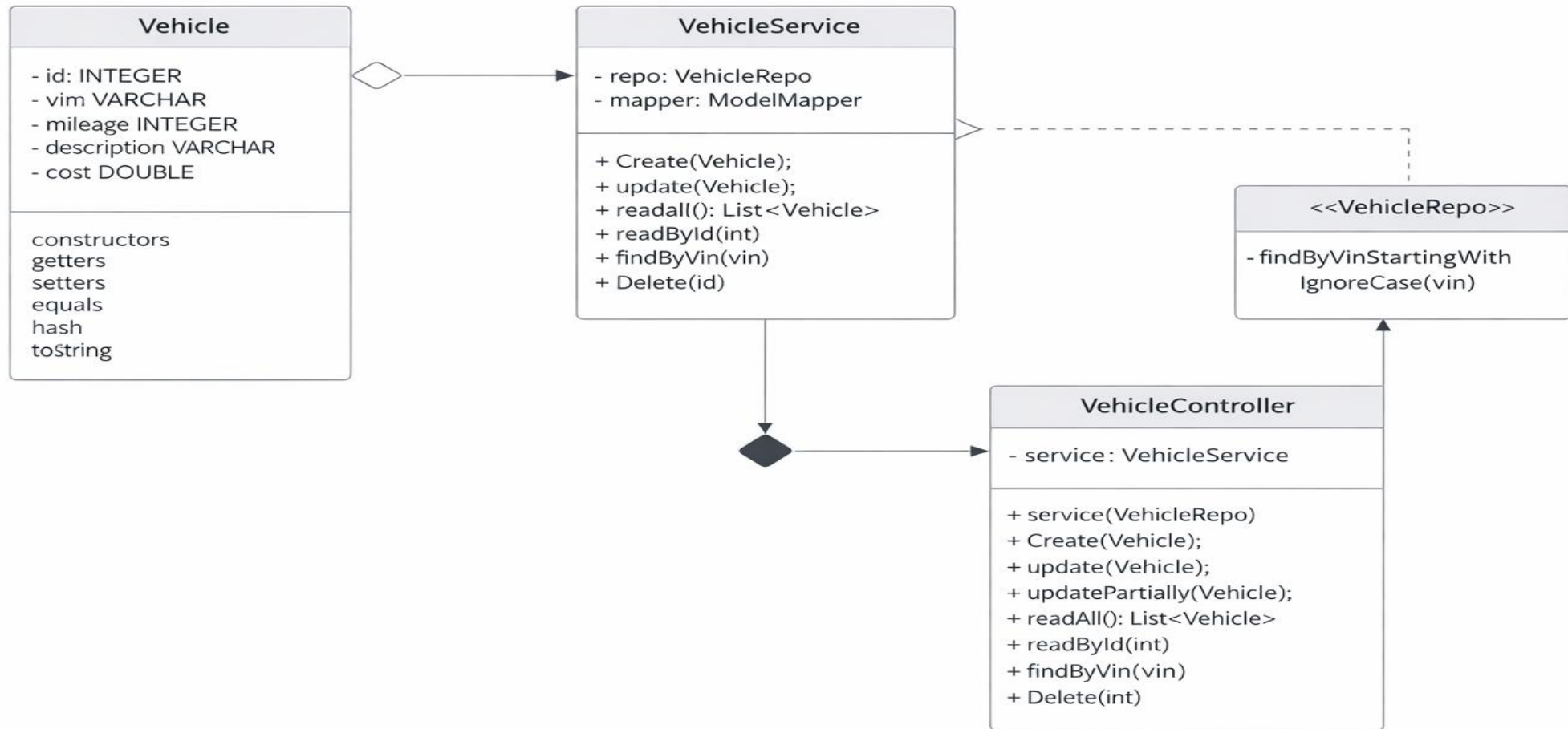
SERVICE BOOKING MODULE

- The Service Booking Module is the core functional module of the Vehicle Service Management System. It enables users to book service appointments for their registered vehicles in a simple and user-friendly manner. Through this module, users can select a vehicle, choose the type of service, specify the preferred service date, and submit a service request.
- Once a service booking is submitted, the booking status is initially marked as **Pending**. The administrator reviews the request and updates the status to **Approved** or **Completed** based on the service progress. Users can track the real-time status of their bookings through the system. This module ensures transparency, efficient communication between users and administrators, and smooth handling of vehicle service operations.

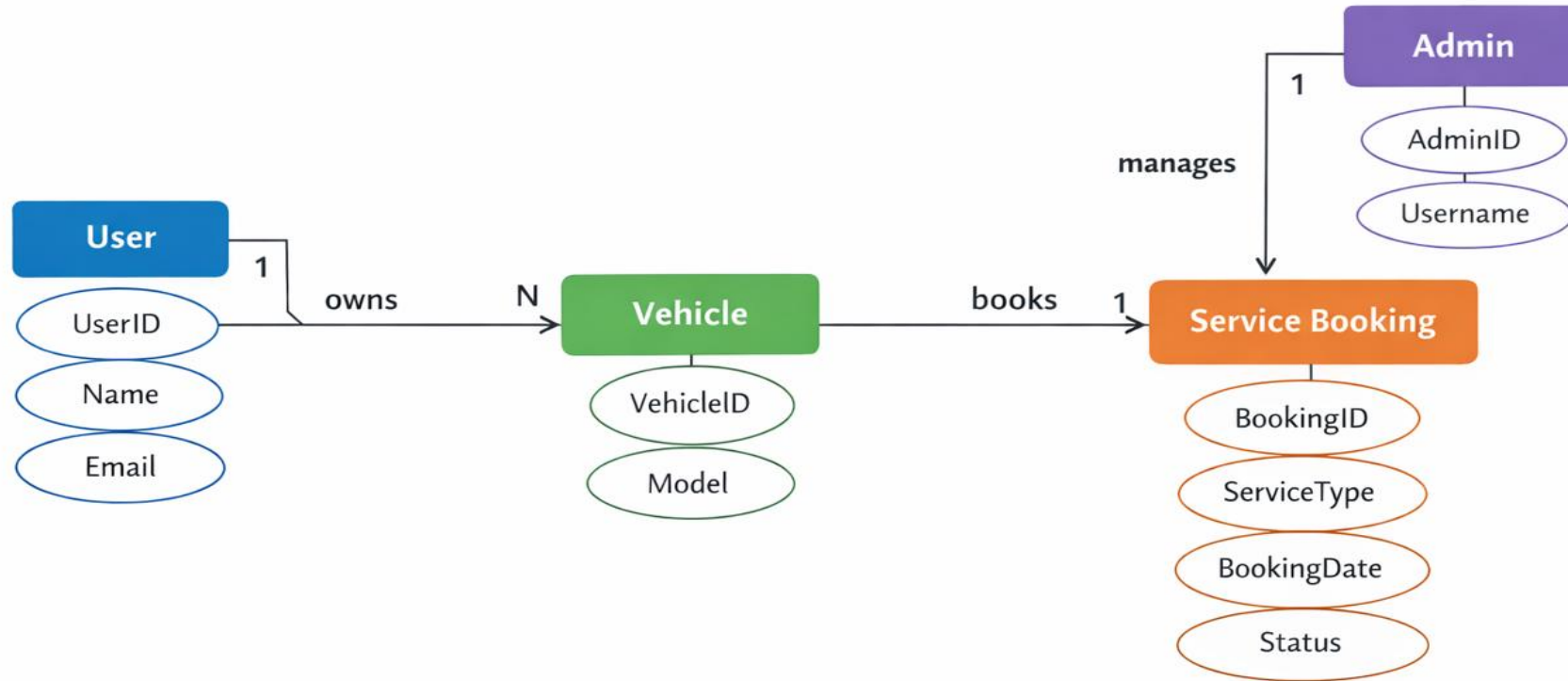
CLIENT SERVER ARCHITECTURE



UML DAIGRAM



Vehicle Service Management System - ER Diagram



DATA DICTIONARY

TABLES OF DATABASES

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration Schemas

Information

No object selected

SQL File 3* SQL File 3* x

```
1 • CREATE DATABASE vehicleservice;  
2 • USE vehicleservice;  
3 • SELECT * FROM vehicle;  
4 • DESC vehicle;  
5 • ALTER TABLE vehicle  
6   MODIFY id BIGINT NOT NULL AUTO_INCREMENT;  
7 • SHOW PROCESSLIST;  
8 • SHOW VARIABLES LIKE 'max_connections';
```

Limit to 1000 rows

Result Grid

	price	id	owner_name	vehicle_number	vehicle_type
▶	0	1	Normal User	TN2W300	Auto-Added
*	HULL	HULL	HULL	HULL	HULL

vehicle 1 x

Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	20:19:16	USE vehicleservice	0 row(s) affected	0.000 sec
✓ 2	20:19:20	SELECT * FROM vehicle LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid
Form Editor
Field Types
Query Stats

20:19 17-12-2025

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

SQL File 3* SQL File 3* x

Limit to 1000 rows

MANAGEMENT

Server Status
Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore

INSTANCE

Startup / Shutdown
Server Logs

Administration Schemas

No object selected

SQL File 3*

1 • CREATE DATABASE vehicleservice;
2 • USE vehicleservice;
3 • SELECT * FROM vehicle;
4 • DESC vehicle;
5 • ALTER TABLE vehicle
6 MODIFY id BIGINT NOT NULL AUTO_INCREMENT;
7 • SHOW PROCESSLIST;
8 • SHOW VARIABLES LIKE 'max_connections';

Result Grid

Field	Type	Null	Key	Default	Extra
price	double	YES		NULL	
id	bigint	NO	PRI	NULL	auto_increment
owner_name	varchar(255)	YES		NULL	
vehicle_number	varchar(255)	YES		NULL	
vehicle_type	varchar(255)	YES		NULL	

Result 2 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	20:19:16	USE vehicleservice	0 row(s) affected	0.000 sec
✓ 2	20:19:20	SELECT * FROM vehicle LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓ 3	20:20:43	DESC vehicle	5 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid
Form Editor
Field Types
Query Stats

Context Help Snippets



Search



ADMIN AND USER

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration Schemas

Information

No object selected

SQL File 3*

```
7 • SHOW PROCESSLIST;
8 • SHOW VARIABLES LIKE 'max_connections';
9 • SET GLOBAL max_connections = 200;
10 • SHOW PROCESSLIST;
11 • SELECT email, password FROM users;
12 • DESC users;
13 • INSERT INTO users (email, name, password, role)
14 • VALUES /
```

Result Grid

	email	password
▶	admin@vms.com	\$2a\$10\$bQaEN8oHmXleJ0SoDQ/ouSvQ3MTL4M...
	user@vms.com	\$2a\$10\$mauwLqNfat7s8Gn.BGFVoOEKf0s2ASB...
	veera@vms.com	\$2a\$10\$TMlwWhNkuhJmkWnazAqGOWJ4Llk...

users 4 ×

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 2	20:19:20	SELECT * FROM vehicle LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓ 3	20:20:43	DESC vehicle	5 row(s) returned	0.000 sec / 0.000 sec
✓ 4	20:21:46	SHOW PROCESSLIST	3 row(s) returned	0.016 sec / 0.000 sec
✓ 5	20:21:58	SELECT email, password FROM users LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Read Only

Context Help Snippets

ENG IN 20:22 17-12-2025

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

SQL File 3* SQL File 3* x

Limit to 1000 rows

21 • delete from users;
22
23 • use vehicleservice;
24 • select * from users;
25 • delete from users;
26 • SELECT * FROM service_booking;
27 • CREATE TABLE service_booking (
28 • 14 BTCTMY DDTMADY KEY AUTO INCREMENT

MANAGEMENT

Server Status
Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore

INSTANCE

Startup / Shutdown
Server Logs

Administration Schemas

Information

No object selected

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Content:

	preferred_date	id	user_id	vehicle_id	problem_description	service_type	status
▶	2005-10-09	1	2	1	no	General Service	COMPLETED
	2005-11-09	2	1	1	no	General Service	APPROVED
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

service_booking5 x

Apply Revert Context Help Snippets

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 3	20:20:43	DESC vehicle	5 row(s) returned	0.000 sec / 0.000 sec
✓ 4	20:21:46	SHOW PROCESSLIST	3 row(s) returned	0.016 sec / 0.000 sec
✓ 5	20:21:58	SELECT email, password FROM users LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 6	20:23:45	SELECT * FROM service_booking LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

Object Info Session



Search

ENG
IN20:24
17-12-2025



Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help



Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration Schemas

Information

No object selected

SQL File 3*

SQL File 3* x



Limit to 1000 rows

```
42 VALUES ('Admin', 'admin@gmail.com', '$2a$10$encryptedPassword', 'ROLE_ADMIN');
43 • select * from users;
44 • SELECT email, role FROM users;
45 • show tables;
46
47 • SELECT email, role FROM users;
48 • CREATE TABLE roles (
49     id BIGINT AUTO_INCREMENT PRIMARY KEY,
50     name VARCHAR(50) UNIQUE NOT NULL
51 );
52
53 • CREATE TABLE user_roles (
54     user_id BIGINT NOT NULL,
55     role_id BIGINT NOT NULL,
56     PRIMARY KEY (user_id, role_id),
57     FOREIGN KEY (user_id) REFERENCES users(id),
58     FOREIGN KEY (role_id) REFERENCES roles(id)
59 );
60
61 • INSERT INTO roles (name) VALUES ('ROLE_USER');
62 • INSERT INTO roles (name) VALUES ('ROLE_ADMIN');
63
64
```

PROJECT OUTPUTS

USER REGISTER

Browser tabs: Create Account - VSMS, New tab, receipt_7 (1).pdf

Address bar: localhost:8096/register

Notification: Google Chrome isn't your default browser. Set as default



Create Account

Join Vehicle Service Management System

Full Name

Email Address

Password

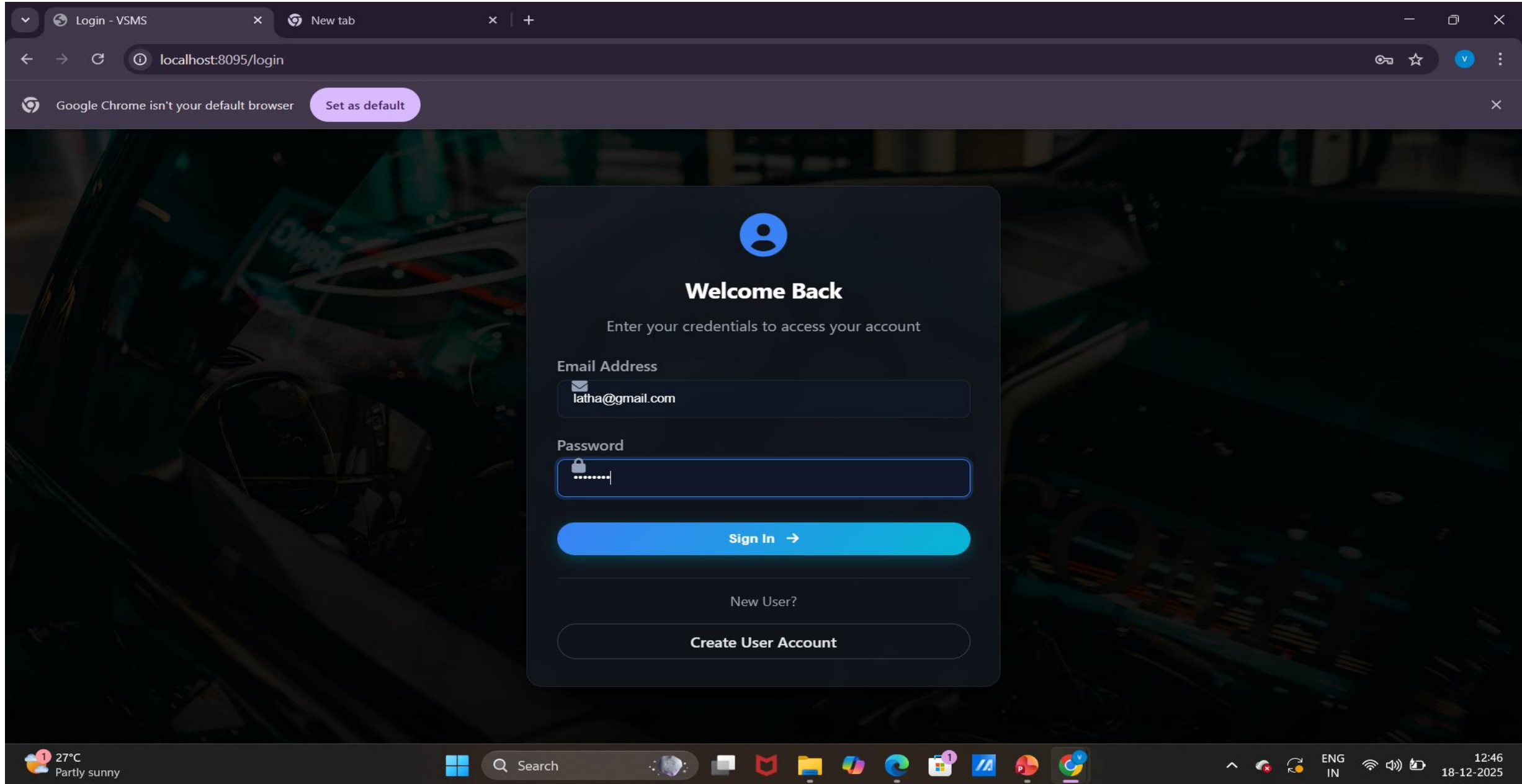
[Register Now](#)

Already have an account?

[Back to Login](#)

Taskbar: 28°C Partly sunny, Search, Task View, Edge, File Explorer, VS Code, Chrome, PowerPoint, Outlook, 13:07 18-12-2025

USER LOGIN PAGE



HOME PAGE

Vehicle Service Management Sy

New tab

localhost:8096/user/dashboard

Google Chrome isn't your default browser

Set as default

VSMS Portal

Home

My Vehicles

Add Vehicle

Book Service

My Bookings

Logout

Vehicle Service Management System

Manage vehicle details, owners and service information in a simple and efficient way.

View Vehicles

Add Vehicle

27°C

Partly sunny

Search

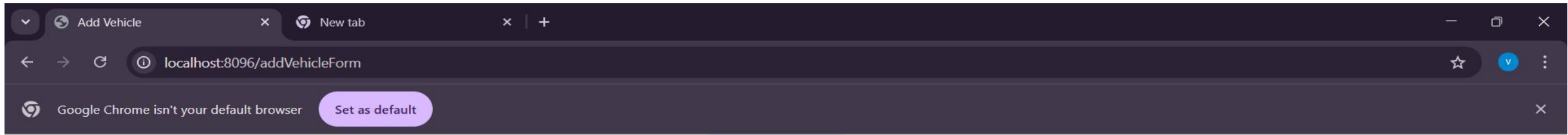
ENG

IN

12:50

18-12-2025

ADD VEHICLE PAGE



Add / Edit Vehicle

Vehicle Number:

TN05CQ1656

Vehicle Type:

Bike

Owner Name:

Shawn mike

Save Vehicle

Cancel

BOOK SERVICE PAGE

Book ServiceNew tab

localhost:8096/bookService

Google Chrome isn't your default browserSet as default

VSMS Portal

HomeMy VehiclesAdd VehicleBook ServiceMy BookingsLogout

Book Vehicle Service

Select Your Vehicle

TN05CQ1656 (Bike)

Vehicle Number

TN05CQ1656

Service Price: ₹0.00

(Fixed by Admin)

Service Type

Engine Check

Preferred Date

25-12-2025

Problem Description

exhaust retunning

28°CPartly sunny

Search

ENG IN

12:5718-12-2025

TN05CQ1656

Service Price: ₹0.00
(Fixed by Admin)

Service Type

Engine Check

Preferred Date

25 - 12 - 2025

Problem Description

exhaust retunning

Book Service

Cancel

My Bookings

New tab

localhost:8096/myBookings

Google Chrome isn't your default browser

Set as default

VSMS Portal

Home

My Vehicles

Add Vehicle

Book Service

My Bookings

Logout

TOTAL BOOKINGS

3

PENDING

1

APPROVED

1

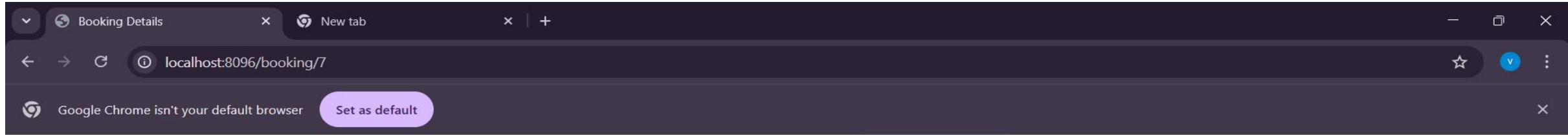
COMPLETED

1

My Bookings

VEHICLE NO	SERVICE TYPE	DATE	PRICE	STATUS
TN08AB1234	Oil Change	18 Dec 2025	Price Pending	PENDING
TN09AB89123	General Service	18 Dec 2025	\$3500.0	COMPLETED

BOOKING DEATILS



Booking Details

Status: **COMPLETED**

Service Date: 2025-12-18

Service Price: **\$3500.0**

[← Back to My Bookings](#)

 **Download Receipt**

RECEIPT

Booking Details

receipt_7 (1).pdf

New tab

File

C:/Users/sabit/Downloads/receipt_7%20(1).pdf

☆

📄

🔍

Google Chrome isn't your default browser

Set as default

receipt_7 (1).pdf

1 / 1 | 100% | 🖨️ 🔍 🔒 ⏪ ⏩

📄 🖨️ ⋮

Vehicle Service Receipt

Booking ID	7
Service Date	2025-12-18
Booking Status	COMPLETED
Vehicle Number	TN09AB89123
Service Type	General Service
Service Price	\$3500.0

1

28°C
Partly sunny

🔍 Search

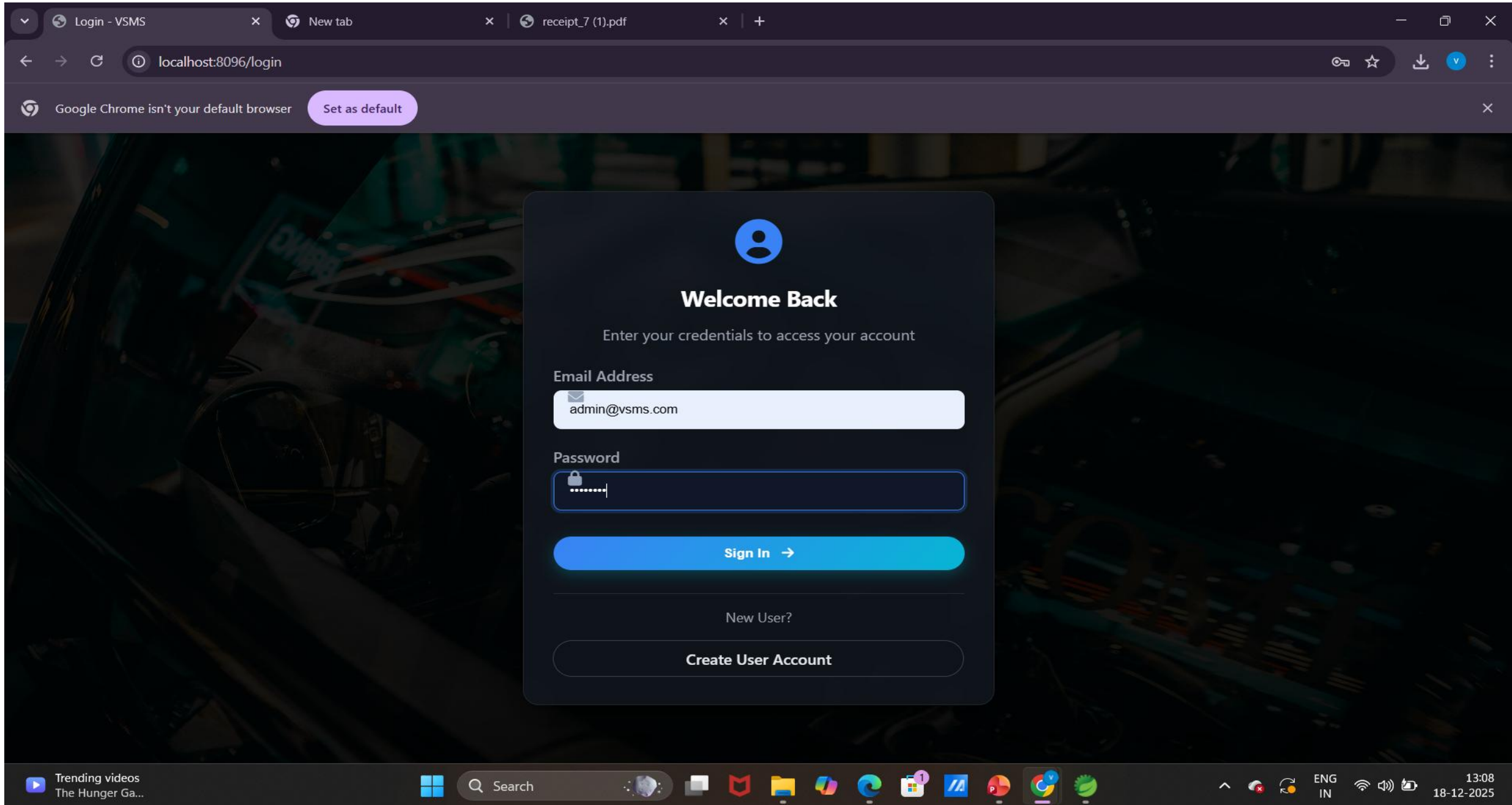
🖨️ 📧 📁 🎨 🌐 📅 📄 📌 📎

ENG
IN

📶 🔊 🔌

13:03
18-12-2025

ADMIN LOGIN PAGE



ADMIN DASHBOARD

The screenshot shows a web browser window with the following details:

- Browser Tabs:** Admin Dashboard, New tab, receipt_7 (1).pdf
- Address Bar:** localhost:8096/admin/dashboard
- Notification:** Google Chrome isn't your default browser (Set as default)
- Dashboard Header:** Admin Dashboard (left), Dashboard Vehicles Bookings Users Logout (right)
- Dashboard Content:** Five cards showing statistics: TOTAL USERS (8), TOTAL VEHICLES (7), TOTAL BOOKINGS (9), PENDING (3), and APPROVED (3). A sixth card, COMPLETED (3), is partially visible on the left.
- Taskbar:** 27°C Partly sunny, Search bar, and various application icons (File Explorer, Edge, etc.).
- System Tray:** ENG IN, network icons, and date/time: 13:10 18-12-2025.

Category	Count
TOTAL USERS	8
TOTAL VEHICLES	7
TOTAL BOOKINGS	9
PENDING	3
APPROVED	3
COMPLETED	3

MANAGE VEHICLES

Manage Vehicles

New tab

receipt_7 (1).pdf

localhost:8096/admin/vehicles

Google Chrome isn't your default browser

Set as default

Manage Vehicles

DashboardVehiclesBookingsUsers

ID	VEHICLE NUMBER	TYPE	OWNER	ACTIONS
1	TN2W300	Auto-Added	Normal User	Delete
3	TN09AB1234	Car	Muruganatham	Delete
4	TN09AB1234	Car	Muruganatham	Delete
5	TN09AB5678	Bike	priya	Delete
6	TN08AB1234	royalenfield	latha	Delete
7	TN09AB89123	car	latha	Delete
8	TN05CQ1656	Bike	Shawn mike	Delete

27°C
Partly sunny

Search

ENG
IN

13:11
18-12-2025

MANAGE BOOKINGS

Manage Bookings

New tab

receipt_7 (1).pdf

localhost:8096/admin/bookings

Google Chrome isn't your default browser

Set as default

Manage Bookings

DashboardVehiclesBookingsUsers

ID	STATUS	SERVICE TYPE	DATE	PRICE	ACTIONS
1	COMPLETED	General Service	2005-10-09	\$2000.0	<button>View</button>
2	APPROVED	General Service	2005-11-09	Not Set	<button>View</button> <button>Complete</button>
3	APPROVED	General Service	2025-11-09	Not Set	<button>View</button> <button>Complete</button>
4	PENDING	Brake Service	2025-12-17	Not Set	<button>View</button> <button>Approve</button>
5	PENDING	Brake Service	2025-12-17	Not Set	<button>View</button> <button>Approve</button>
6	PENDING	Oil Change	2025-12-18	Not Set	<button>View</button> <button>Approve</button>
7	COMPLETED	General Service	2025-12-18	\$3500.0	<button>View</button>

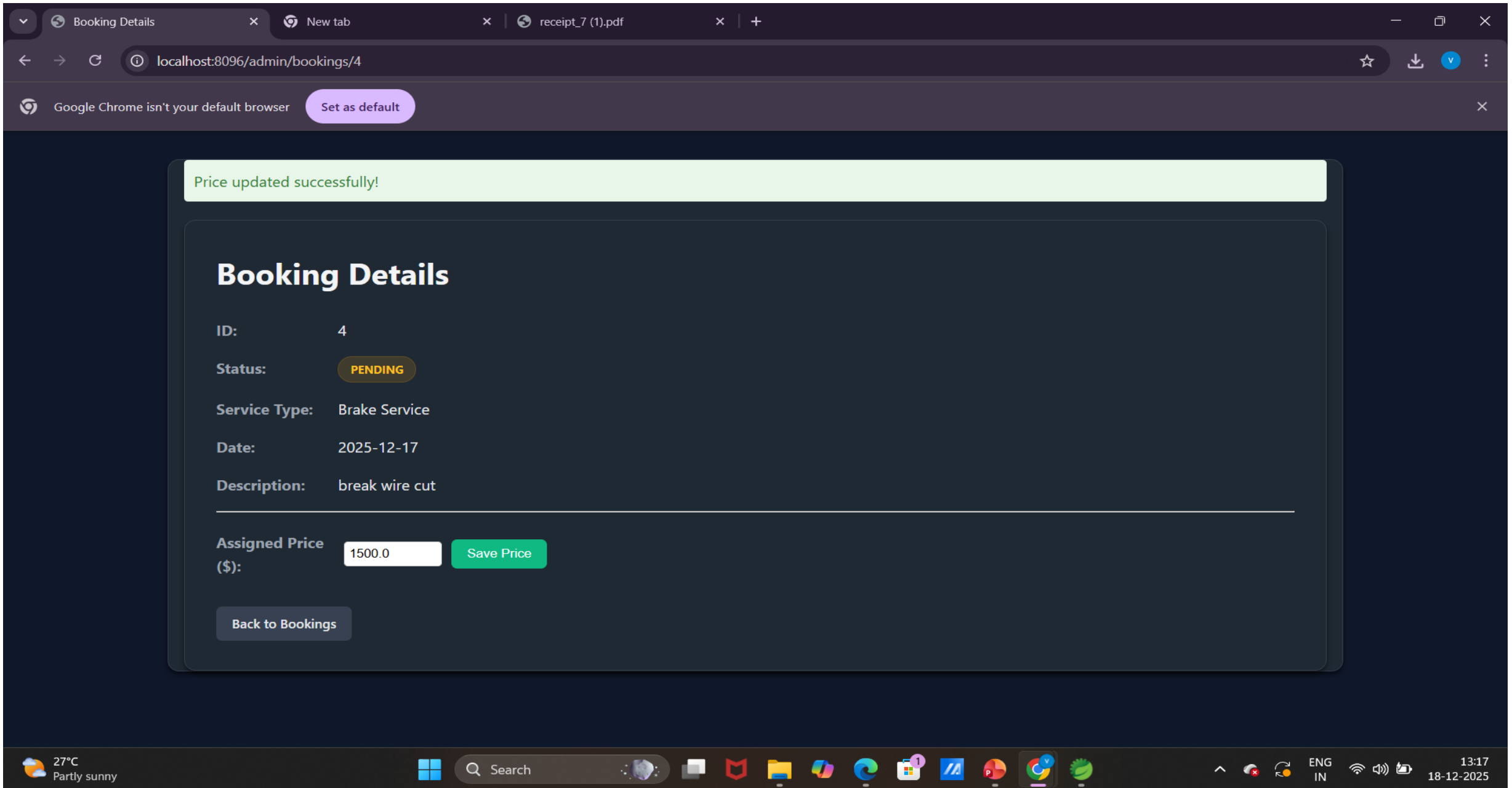
localhost:8096/admin/bookings

27°C
Partly sunny

Search

ENG
IN

13:13
18-12-2025



MANAGE USER

Manage Users

New tab

receipt_7 (1).pdf

localhost:8096/admin/users

Google Chrome isn't your default browser

Set as default

Manage Users

Dashboard

Vehicles

Bookings

Users

ID	NAME	EMAIL	ROLE	ACTIONS
1	Admin User	admin@vsms.com	ROLE_ADMIN	Delete
2	Normal User	user@vsms.com	ROLE_USER	Delete
4	veera	veera@vsms.com	ROLE_USER	Delete
5	Sabitha Anandh	sabithaanandh@vsms.com	ROLE_USER	Delete
6	test2	test2@gmail.com	ROLE_USER	Delete
7	ravanan	ravanan@gmail.com	ROLE_USER	Delete
8	priya	priya@gmail.com	ROLE_USER	Delete
9	latha	latha@gmail.com	ROLE_USER	Delete

27°C

Partly sunny

Search

ENG IN

13:15

18-12-2025

VIDEO/GITHUB LINK

- GITHUB: https://github.com/sabithanandhm/sabitha-Anandh_4373110
- DEMO VIDEO : <https://drive.google.com/file/d/1oyUsp0cQE3JCFUZ8E-l3zDG7lOqvZyJQ/view?usp=drives>

THANK YOU